

ST ANTHONY, NL, Canada

WMO: 718194

Lat: 51.370N

Lon: 55.600W

Elev: 38

StdP: 14.68

Time Zone: -3.50 (NAF)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.0	-11.2	-24.7	1.4	-14.5	-20.6	1.8	-10.6	41.2	19.3	37.6	22.0	16.6	270	0.962

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.4	69.0	60.7	65.8	59.0	62.9	57.5	62.3	66.7	60.2	64.1	58.2	61.9	17.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.6	79.0	63.7	58.2	72.6	62.4	56.3	67.6	60.8	28.2	66.6	26.8	64.0	25.4	62.0	69.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.1	29.6	25.8	DB	-20.0	76.5	2.7	2.4	-21.9	78.3	-23.4	79.7	-24.9	81.1	-26.9	82.8	(4)
(5)			WB	-20.4	66.7	2.6	2.3	-22.3	68.4	-23.9	69.7	-25.4	71.0	-27.3	72.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.2	9.7	8.8	18.3	28.6	36.4	44.0	51.3	54.5	47.8	38.6	28.2	18.2		
	DBStd	17.07	11.06	11.09	9.68	5.98	6.11	6.46	6.58	5.65	5.19	5.46	7.35	9.84		
	HDD50	6837	1250	1153	981	642	426	202	65	20	104	353	654	987		
	HDD65	11986	1715	1573	1446	1092	888	630	426	327	516	817	1104	1452		
	CDD50	330	0	0	0	0	3	22	106	160	38	1	0	0		
	CDD65	4	0	0	0	0	0	1	1	2	0	0	0	0		
	CDH74	9	0	0	0	0	0	3	2	4	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	14.5	17.4	17.2	16.8	14.5	13.3	12.5	11.7	12.2	13.2	13.4	15.5	16.4		
	PrecAvg	42.5	4.4	3.7	3.2	2.4	2.8	3.0	3.5	3.7	3.7	3.9	4.1	4.3		
(15) Precipitation	PrecMax	51.3	9.1	5.8	5.7	4.2	5.7	6.0	7.0	7.6	7.8	7.3	7.8	6.3		
	PrecMin	35.2	1.8	1.9	1.3	0.8	0.7	1.2	1.4	1.1	1.3	1.6	1.7	2.0		
	PrecStd	3.9	1.6	1.1	1.1	0.8	1.2	1.0	1.2	1.3	1.2	1.4	1.3	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	35.6	38.8	45.7	61.5	69.3	72.8	74.4	65.1	55.5	46.6	36.5		
		MCWB	36.9	35.4	36.3	42.3	48.3	58.8	62.9	64.5	57.7	51.4	45.9	36.2		
	2%	DB	32.0	32.6	35.7	40.9	53.7	62.1	68.8	69.7	61.5	51.3	42.3	33.8		
		MCWB	31.4	31.9	34.9	37.6	45.6	55.2	60.7	61.6	55.7	49.3	41.2	32.8		
	5%	DB	29.3	29.7	33.6	38.2	49.5	58.8	65.6	66.3	58.7	48.9	39.2	32.3		
		MCWB	28.5	28.9	32.7	35.3	43.8	52.9	58.8	60.1	53.7	46.9	37.9	31.7		
	10%	DB	26.5	25.9	32.0	35.7	45.8	55.5	62.7	63.4	56.1	46.5	37.2	30.9		
		MCWB	25.7	24.9	31.2	33.5	41.7	50.6	57.2	58.4	52.2	44.4	36.1	30.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.1	35.5	37.0	42.8	50.5	60.5	65.9	66.0	59.3	53.4	46.3	36.3		
		MCDB	37.1	35.5	37.8	45.8	57.0	67.2	68.6	71.4	63.2	54.3	46.6	36.4		
	2%	WB	31.4	32.1	35.0	38.6	47.1	57.1	62.3	62.9	56.9	50.0	41.6	33.2		
		MCDB	31.7	32.5	35.7	40.0	52.1	60.5	67.0	67.1	60.3	50.9	42.0	33.4		
	5%	WB	28.6	28.8	32.9	35.6	44.4	54.2	60.1	60.8	54.7	47.3	38.3	31.8		
		MCDB	29.2	29.4	33.6	37.5	48.6	57.5	64.4	64.4	57.7	48.4	38.9	32.2		
	10%	WB	25.6	24.9	31.3	33.8	41.9	51.0	57.8	59.0	52.8	44.7	36.3	30.1		
		MCDB	26.4	25.8	32.0	35.2	45.7	55.1	61.9	62.2	55.3	46.2	37.0	30.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	13.2	12.3	8.6	11.0	13.7	12.8	12.4	11.7	9.2	9.1	11.6		
		MCDBR	17.8	14.2	11.9	11.1	18.3	20.5	18.8	16.2	16.0	12.2	10.0	11.2		
		MCWBR	18.2	14.3	11.9	8.8	12.1	14.4	11.5	9.8	11.2	10.7	10.3	11.7		
	5% WB	MCDBR	18.1	13.7	11.9	9.9	16.8	19.8	16.7	14.7	14.5	11.3	10.0	11.7		
		MCWBR	18.7	14.0	12.1	8.3	12.5	15.3	12.0	11.3	11.4	11.4	10.7	12.2		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.289	0.308	0.349	0.373	0.391	0.406	0.379	0.358	0.327	0.299	0.290		
	taud		2.320	2.283	2.270	2.260	2.349	2.364	2.372	2.443	2.478	2.521	2.478	2.316		
	Ebn at Noon		233	266	282	282	279	274	268	270	264	252	230	208		
	Edh at Noon		20	27	34	38	37	37	36	32	28	22	17	17		
(44) All-Sky Solar Radiation	RadAvg		252	551	955	1331	1521	1598	1584	1334	990	586	306	193		
	RadStd		32	75	137	153	176	140	137	84	80	53	28	23		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	+2.59	N/A	+0.67	+0.99	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page